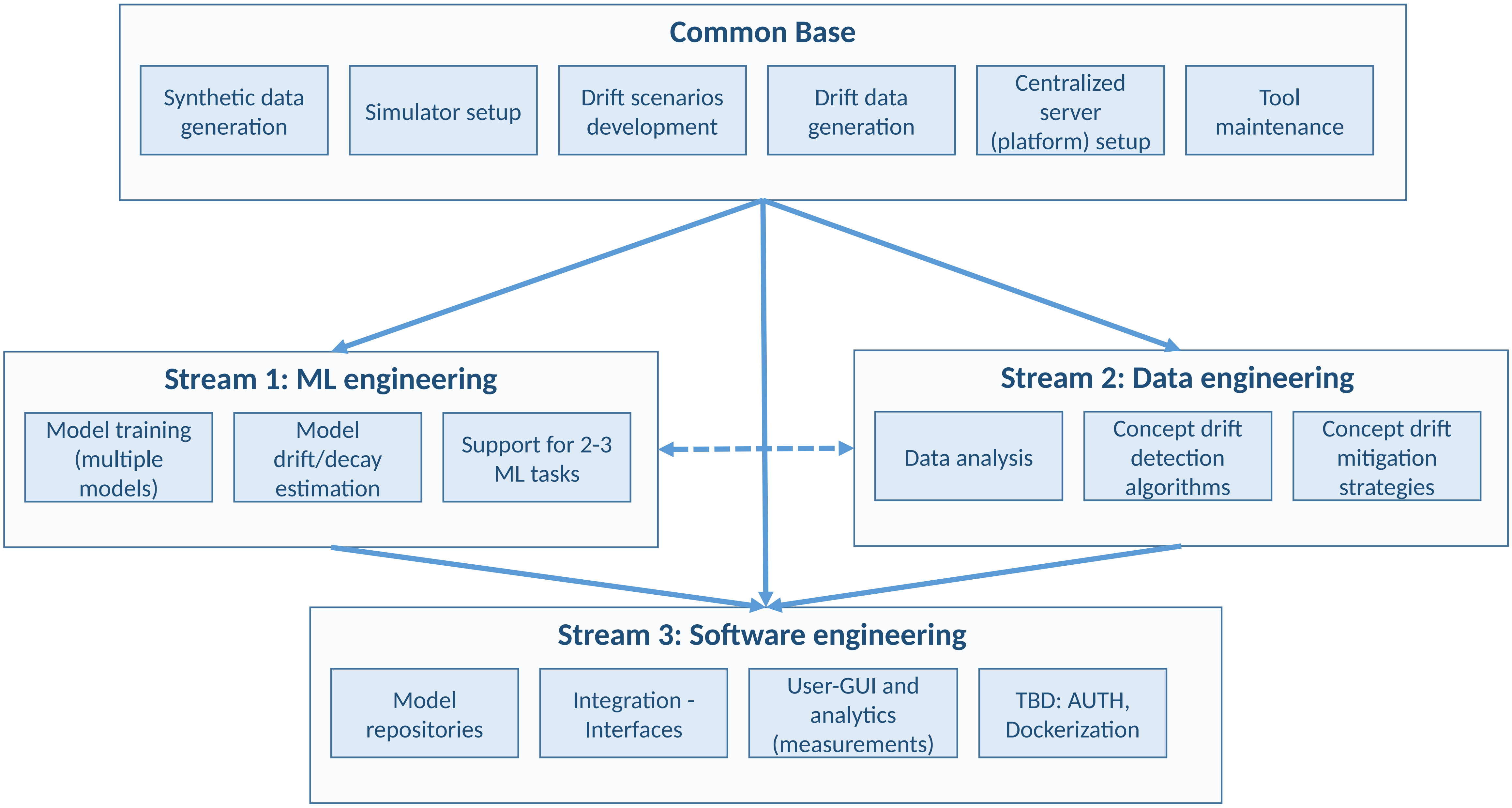
Thesis overview

- Title: Continuous Machine Learning under Concept Drift for CCAM applications
- Estimated duration: 6 months – 24 weeks
- Period: 1 Mar – 1 Sep (TBD on final presentation)
- Technical Description:
    - <u>CCAM data simulation</u>: Simulate a connected vehicular environment, where client-vehicles move around an area constantly collecting kinematic and network (synthetic) data.
    - <u>ML model development</u>: Using the synthetic datasets train prediction models to be used as enablers in CCAM application e.g., network delay prediction, vehicle trajectory prediction, etc.
    - <u>Drift scenarios simulation</u>: Manipulate the simulator to include concept drift scenarios – data distribution changes due to an external cause e.g., installation of a new basestation, a change in user mobility patterns/habits.
    - <u>Concept drift detection</u>: Explore statistical tests and methods to accurately and timely detect such drifts in CCAM environments. Implement a decision system to suggest when mitigation actions are needed.
    - <u>Concept drift adaptation</u>: Explore ideas of how to deal with drifted data. Monitor model performance/degradation. Suggest counter-measures for drifted models e.g., re-training, model switch, train from scratch, etc.
    - <u>MLOps platform</u>: Implement an integrated platform that manages all the above-mentioned functionalities centrally. Data and process visualization to the end-user.

Thesis streams



Suggested bibliography and tools (1)

Stream	Item	Sources
Common	Simulator/Synthetic data generation	OMNET/Veins/SUMO/Simu5G (http://simu5g.org/)
Common	Drift scenarios	Lu, Jie, et al. Learning under concept drift: A review.
Common	Data analysis (timeseries)	Brownlee J. Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python
Common	Centralized server setup	Basic Linux cmds (https://www.geeksforgeeks.org/linux-tutorial/)
Stream 1	Model training	Pytorch/LSTM https://pytorch.org/docs/stable/generated/torch.nn.LSTM.html
		XGBoost https://xgboost.readthedocs.io/en/stable/
		ARIMA https://www.kdnuggets.com/2023/08/times-series-analysis-arima-models-python.html
Stream 1	Model drift/decay – Estimators	Bayram, F.,(2022). From concept drift to model degradation: An overview on performance-aware drift detectors
Stream 1	ML tasks	Ip, André, Luis Irio, and Rodolfo Oliveira. "Vehicle trajectory prediction based on LSTM recurrent neural networks." 2021 IEEE 93rd Vehicular Technology Conference (VTC2021-Spring). IEEE, 2021.
		Barmpounakis, Sokratis, et al. "LSTM-based QoS prediction for 5G-enabled Connected and Automated Mobility applications." 2021 IEEE 4th 5G World Forum (5GWF). IEEE, 2021.
		Abduljabbar, Rusul L., Hussein Dia, and Pei-Wei Tsai. "Unidirectional and bidirectional LSTM models for short-term traffic prediction." Journal of Advanced Transportation 2021 (2021): 1-16.

Suggested bibliography and tools (2)

Stream	Item	Sources
Stream 2	Data analysis (preprocessing, cleaning, stat extraction, feature engineering, etc.)	Pandas (https://pandas.pydata.org/)
		Scipy (https://scipy.org/)
		Brownlee J. Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python
Stream 2	Concept drift detection	Abdu, Nail Adeeb Ali, and Khaled Omer Basulaim. "A review of tracking concept drift detection in machine learning." Recent Trends in Computational Sciences (2023): 36-41.
Stream 2	Concept drift mitigation	Hoens, T. Ryan, Robi Polikar, and Nitesh V. Chawla. "Learning from streaming data with concept drift and imbalance: an overview." Progress in Artificial Intelligence 1 (2012): 89-101.
		Lima, Marília, et al. "Learning Under Concept Drift for Regression—A Systematic Literature Review." IEEE Access 10 (2022): 45410-45429.
		Bayram, F.,(2022). From concept drift to model degradation: An overview on performance-aware drift detectors
		Mehmood, Hassan, et al. "Concept drift adaptation techniques in distributed environment for real-world data streams." Smart Cities 4.1 (2021): 349-371.

Suggested bibliography and tools (3)

Stream	Item	Sources
Stream 3	Integration, Interfaces	Flask (https://flask.palletsprojects.com/en/3.0.x/), FastAPI (https://fastapi.tiangolo.com/), Swagger (https://swagger.io/)
		SQL or similar DB (https://dev.mysql.com/doc/mysql-getting-started/en/)
		Linux bash (preliminaries only)
Stream 3	Visualization/GUI	Dash (https://dash.plotly.com/), Grafana (https://grafana.com/)
Stream 3	Repository management and MLOps	Git LFS (https://git-lfs.com/)
		Mlflow (https://mlflow.org/), Kedro (https://github.com/kedro-org/kedro), Kubeflow (https://www.kubeflow.org/)
		Google MLOps: https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning
		Amazon MLOps: https://aws.amazon.com/sagemaker/mlops/?sagemaker-data-wrangler-whats-new.sort-by=item.additionalFields.postDateTime&sagemaker-data-wrangler-
Stream 3	Virtualization (TBD)	Docker: https://www.docker.com/
Stream 3	AUTH and User/Account management (TBD)	Keycloak (https://www.keycloak.org/)

Draft time-plan

Week(s)	Common task	Stream 1 task	Stream 2 task	Stream 3 task
W1-W4	Thesis logistics, Common material reading, Design of overall motivation and general storyline			
W5-W11	Simulator setup, data generation, drift scenarios, drift data generation	ML model design, ML task selection w.r.t. to generated data	Drift detection and mitigation techniques selection	System architecture design
W12-W16	Evaluation setup design (overall and per-stream)	ML tuning, training-testing	Drift management implementation	Integration/Backend development and testing
W17-W20	Evaluation, results & debugging			GUI development and testing
W21-W24	Thesis writing and presentation			